Abdul Rehman M. Nasir

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Experience

AI Engineer — Background Agents (backgrounder.dev) **May 2026 – Present**
Remote

- Built credential isolation for AI-agent sandboxes: user API keys, shared pool keys and Claude/ChatGPT tokens sit behind Daytona secrets, swapped in by the egress proxy only for each provider's hosts during a metered turn.
- Published it as the npm package `@background-agents/sandbox-secrets` (hash-named secrets, race-safe create-or-reuse, stale-secret cleanup cron, proxy keepalive across sandbox restarts).
- Built MCP integrations (Smithery, GitHub App, policy-enforcing Gmail server) and usage-based billing with token metering and Stripe credits; 2nd-largest contributor to the platform (316 commits).

Software Engineer — Gitwit **Jun 2025 – Present**
Remote

- Built Gitwit's browser-based AI coding agent (similar to Cursor), enabling natural-language-driven code generation and editing directly in the browser.
- Built and integrated E2B sandbox templates to run code live with real-time preview, streamlining developer workflows.
- Developed GitHub integrations enabling seamless repository connectivity for AI-driven code modifications.

Full Stack AI Engineer — Softject **Aug 2025 – July 2026**
Part-time | Hybrid

- Built AI agents, automation workflows, and RAG-based features (document ingestion, embeddings, vector search, LLM-powered responses) integrated across Next.js, Node.js, Python/FastAPI, and PostgreSQL.
- Previously AI Engineer Intern (Jun–Aug 2025) and Full Stack Developer (Jan–Jun 2025), building web applications with Next.js and full-stack technologies.

Backend Intern — Central Depository Company of Pakistan **Jul 2024 – Sep 2024**
Onsite — Karachi

- Developed REST APIs and backend microservices using Spring Boot, with SQL databases and JWT-based authentication securing inter-service communication.

Projects

EarthLink AI | *Next.js, Python, FastAPI, Tambo AI, Mapbox*
Live: https://earthlink-ai.vercel.app GitHub: https://github.com/abdulrehmann231/earthlink-ai

- Built an AI-native geospatial platform on Tambo AI where an agentic loop orchestrates 14 tools and 6 living UI components, turning plain-English prompts into map actions (proximity search, extremes, comparisons, heatmaps) within a single turn.
- Derived NDVI, land-surface temperature, and greenness metrics from Harmonized Sentinel-2 Level-2A surface reflectance through a FastAPI + Google Earth Engine pipeline, rendered live via Next.js 16 and Mapbox GL.

Draftly | *Next.js, Fastify, Yjs, WebSockets, PostgreSQL*
Live: https://draftly-app.vercel.app GitHub: https://github.com/abdulrehmann231/draftly

- Engineered a real-time collaborative meeting whiteboard achieving sub-50ms synchronization using Yjs CRDTs over WebSockets.
- Added AI intent classification and Groq-powered meeting summaries to turn freeform whiteboard sessions into structured, shareable notes.

Technical Skills

Languages	TypeScript, JavaScript, Python, C#, Java, C++, C
Frameworks	Node.js, Express.js, React, Next.js, FastAPI, Spring Boot, .NET Framework
Libraries & Tools	MCP (Model Context Protocol), Daytona, Tailwind CSS, Bootstrap, PassportJS, NumPy, Pandas, LangChain, Yjs, Docker, Linux, Git
AI & Databases	RAG, HuggingFace, OpenRouter, PostgreSQL, MongoDB, Firebase, Supabase, Pinecone, Mapbox

Education

FAST NUCES Pakistan — B.S. in Computer Science **Expected 2027**
Karachi, Pakistan

- Certification: Full Stack Web Development — Node.js, Express.js, REST APIs, PostgreSQL, PassportJS